

Software Requirements Specification (SRS)

Train Seat Reservation System

1. Introduction

1.1 Purpose

To develop an online system for booking train seats easily and securely.

1.2 Scope

The system allows users to search trains, check seat availability, book tickets, make payments, cancel tickets, and generate e-tickets with PNR.

1.3 Definitions / Acronyms

- **PNR** – Passenger Name Record
- **E-Ticket** – Electronic Ticket
- **Admin** – System Administrator

1.4 References

- IEEE SRS Standard
- Software Engineering Textbook

2. Overall Description

2.1 Product Perspective

Web/mobile-based system connected to railway database and payment gateway.

2.2 Product Features

- User Registration & Login
- Train Search
- Seat Availability
- Ticket Booking
- Online Payment
- Ticket Cancellation
- Admin Management

2.3 User Characteristics

Users must know basic computer/mobile usage.

2.4 Constraints

- Internet required
- Secure payment processing
- Must follow railway rules

3. System Features (Tabular Form)

Feature ID	Feature Name	Description
SF1	User Registration	Users can create an account by providing personal details.
SF2	Login	Users login using username and password.

SF3	Train Search	Search trains by source, destination, and date.
SF4	Seat Availability	View available seats in selected class.
SF5	Ticket Booking	Enter passenger details and book ticket.
SF6	Payment	Pay using UPI, card, or net banking.
SF7	Ticket Cancellation	Cancel booked ticket and receive refund.
SF8	Admin Module	Admin can manage trains, schedules, and bookings.

4. External Interface Requirements

- **User Interface:** Simple and user-friendly forms
- **Software Interface:** Database & Payment Gateway
- **Communication Interface:** Internet with HTTPS

5. Non-Functional Requirements

- Fast response time
- Secure data storage
- 24/7 availability
- Easy maintenance

6. Other Requirements

- Generate PNR and E-ticket
- Send confirmation via SMS/Email
- Maintain booking history